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SCAN ME
FOR A DEMO!

Intro & Motivation

SimStudio combines DAS, camera, and GPS data to track vehicles and detect abnormal traffic events. Since cameras and GPS can be limited by weather, lighting, or coverage, DAS adds continuous road sensing through existing fiber-optic infrastructure.

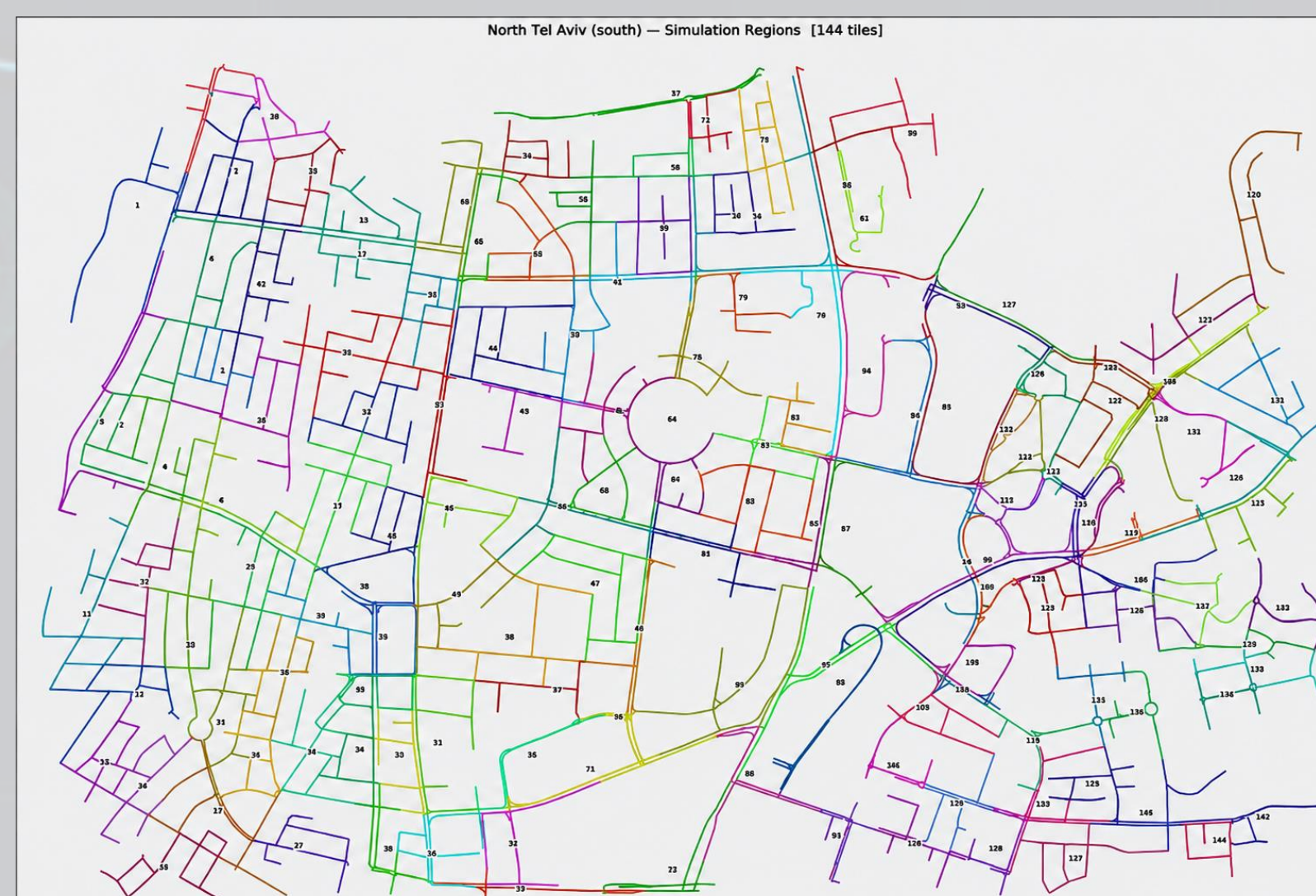
Methods & Implementation

Using SimStudio, we import true street geometry across North Tel Aviv, divided into 144 regions, and generate many anomaly free traffic scenes that set the baseline of normal behavior.

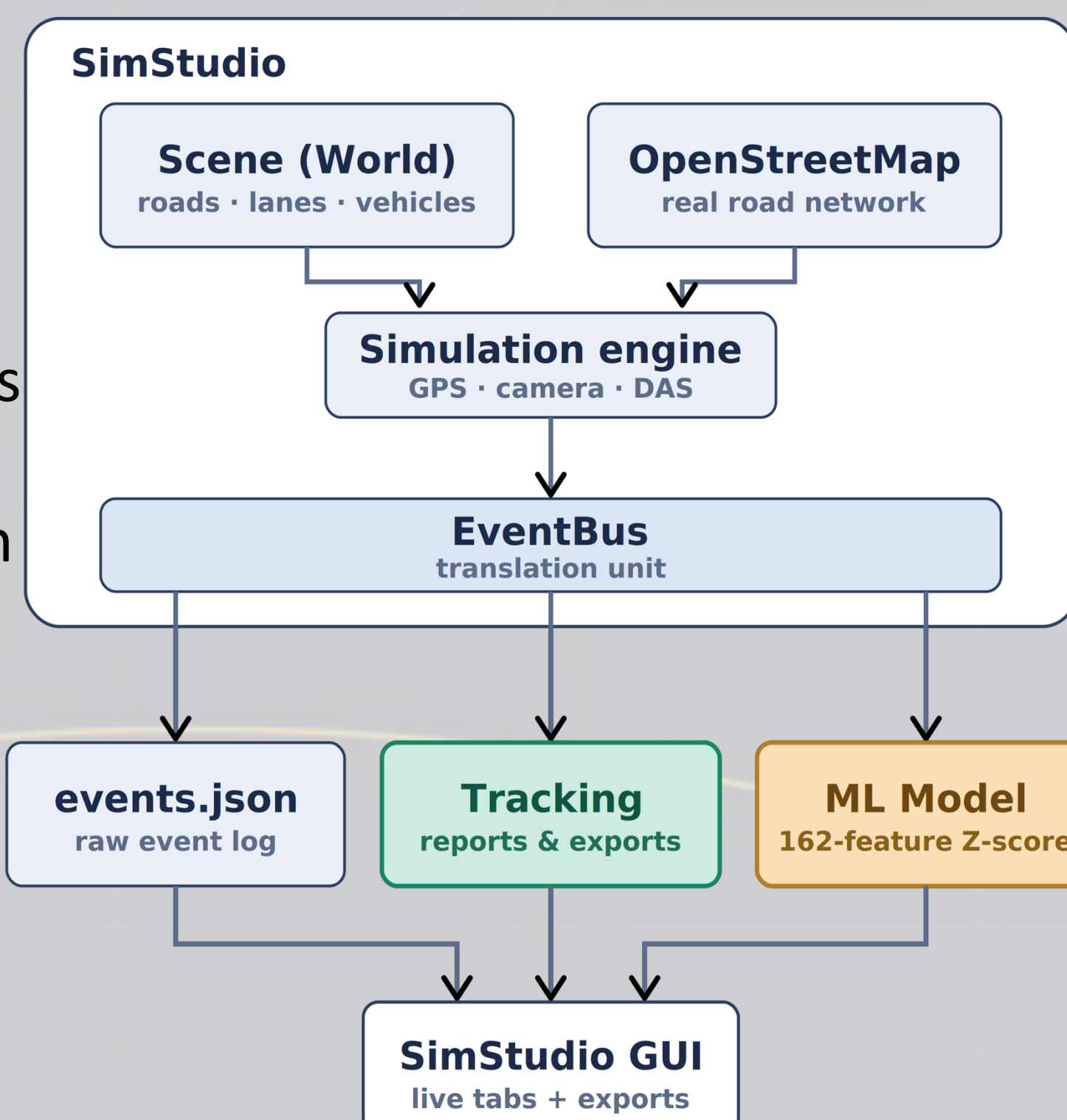
We defined anomalies:

- sudden braking
- Collision
- Speeding
- Stalled Vehicle
- Tailgating
- Heavy Vehicle
- Weaving

and generate multiple scenes to each region. From the resulting.

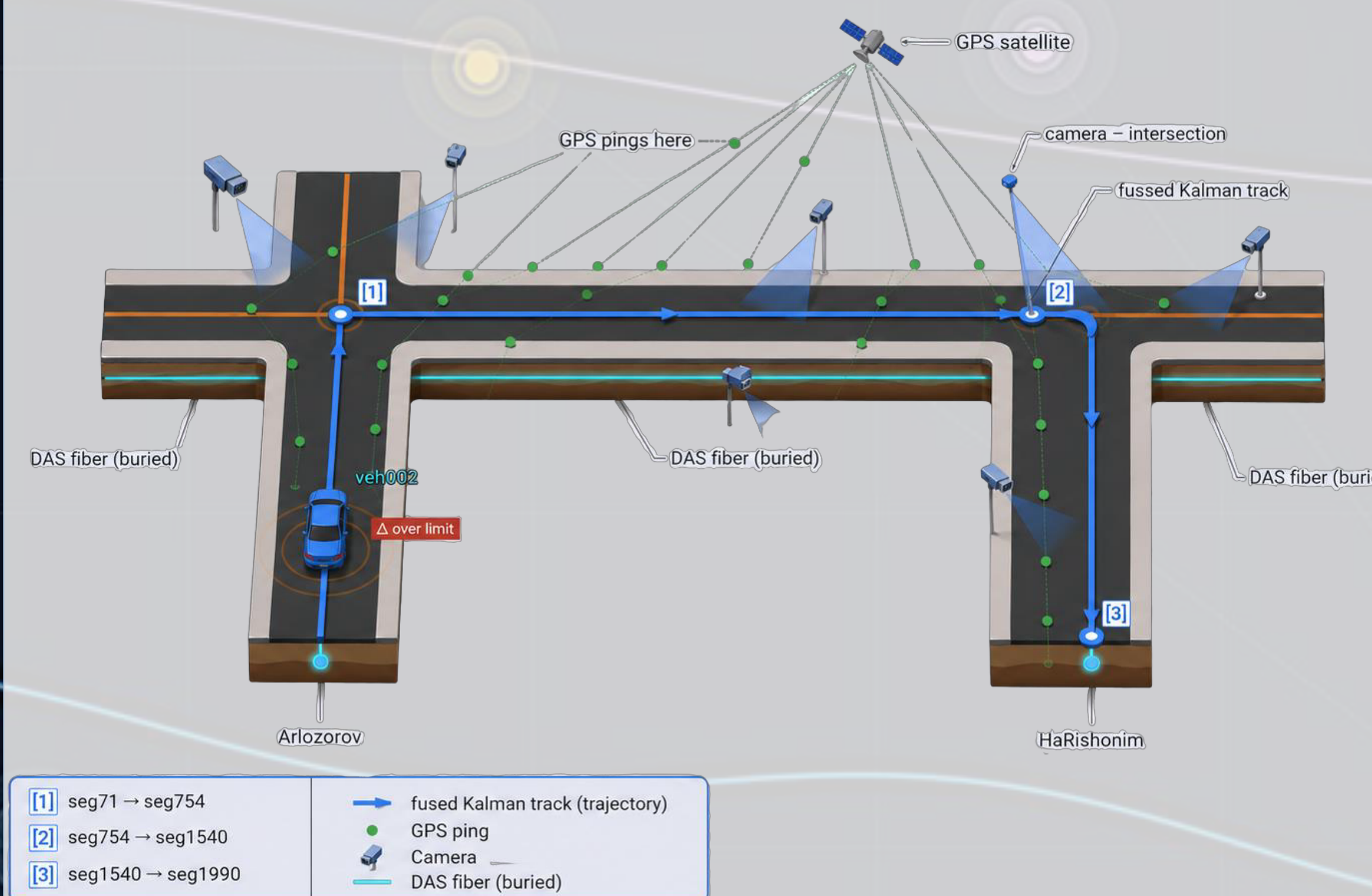


we developed a ML module to identify anomalies. The module learns 162 behavior features for each scene, then scores each vehicle in the scene by how far it deviates - Z score, flagging anything past 15.8σ .

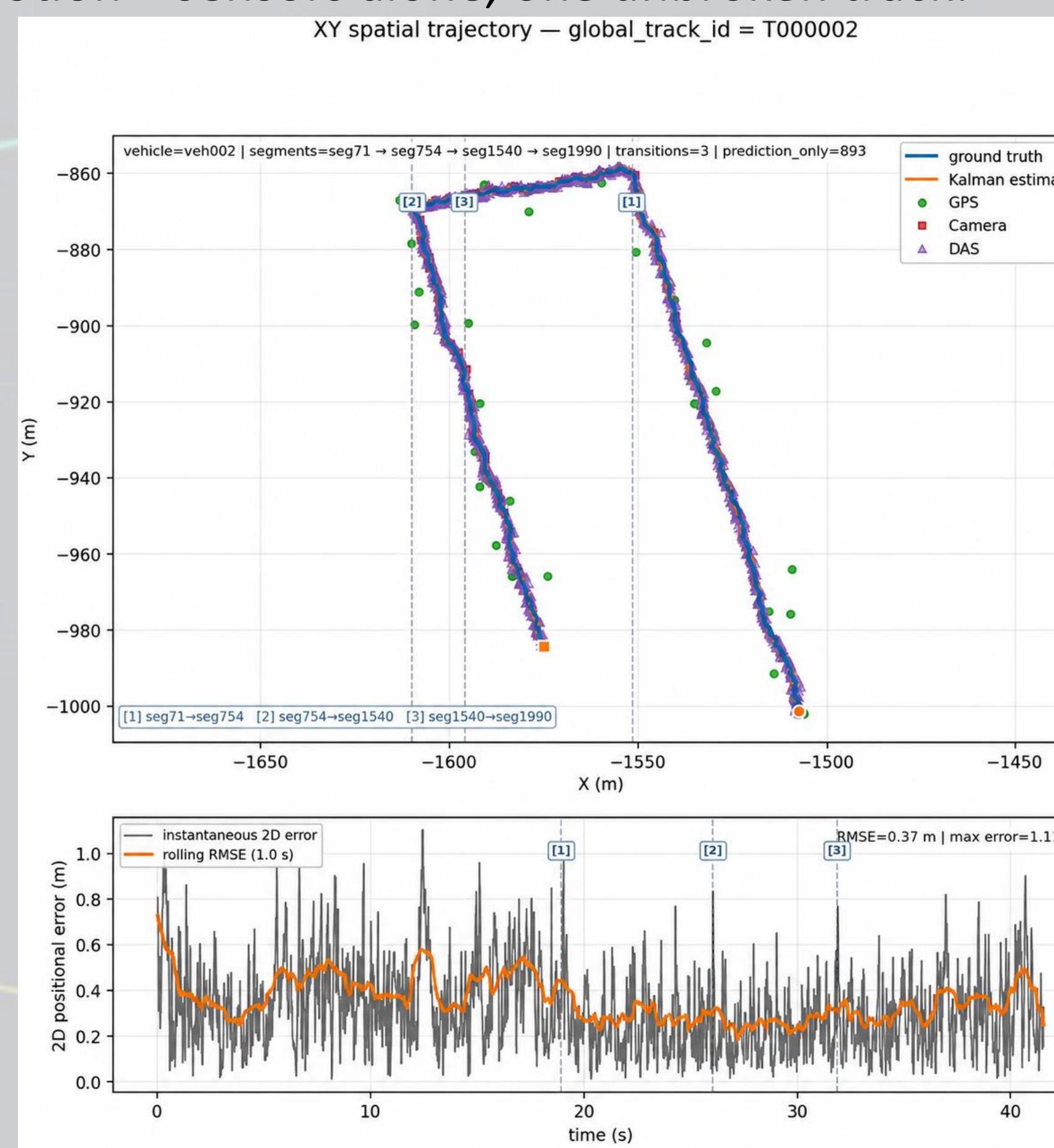


Results - Region 120, over speeding

CROSS SEGMENT TRACKING

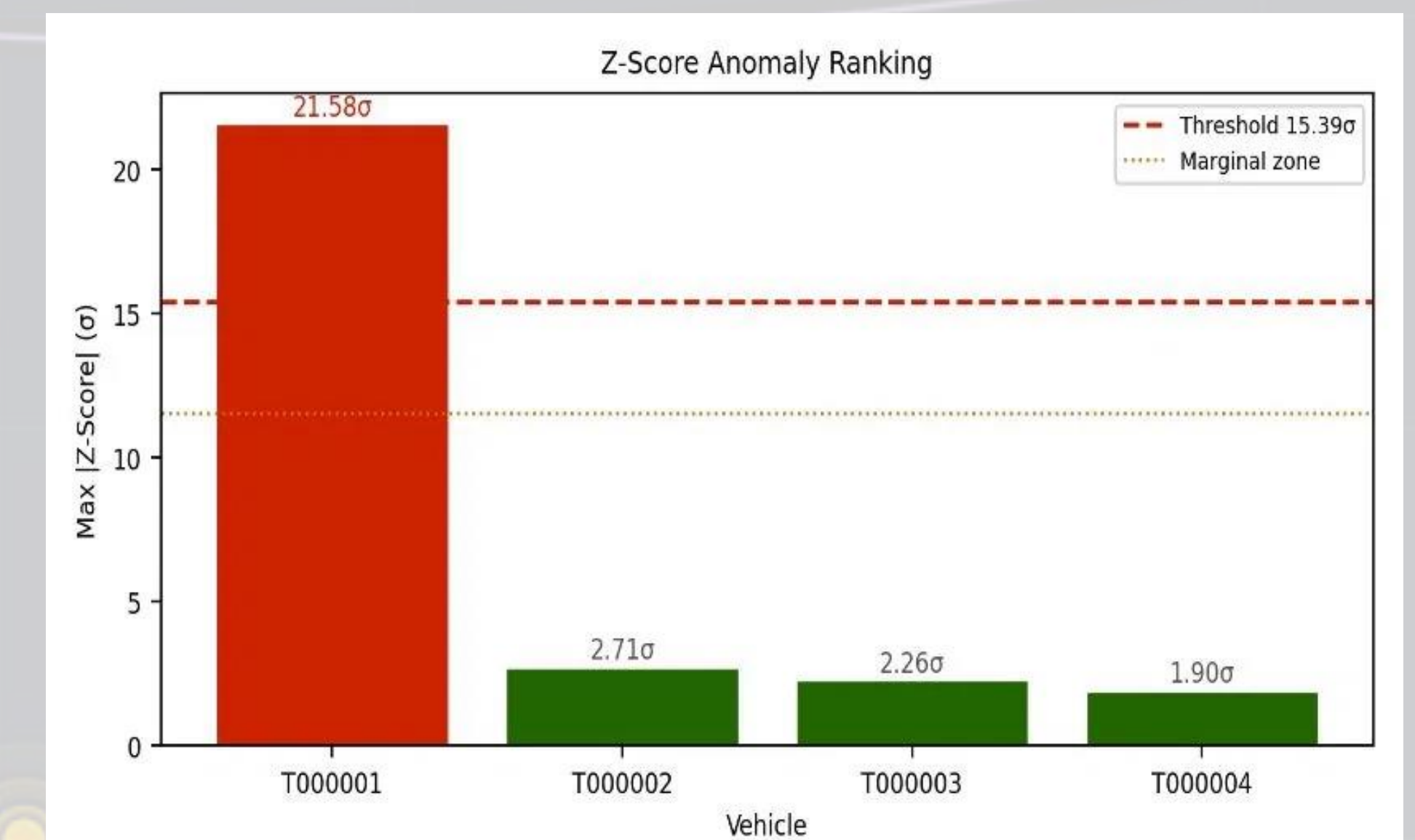


- Kalman filter with 0.37 m RMSE of ground truth.
- Achieving over 90% accuracy and less than 5 km/h speed estimation error.
- Same vehicle re identified across 3 segment from motion + sensors alone, one unbroken track.

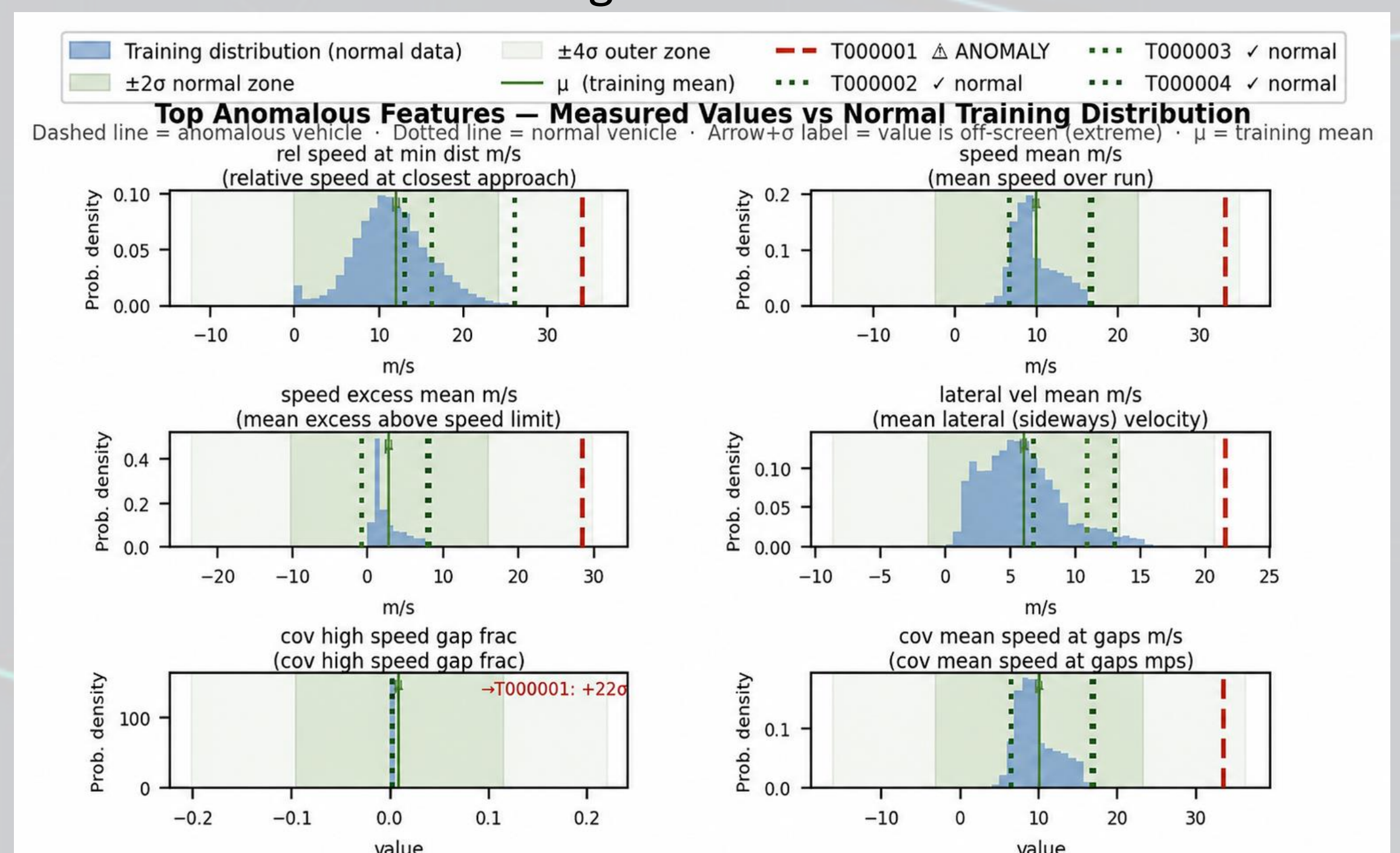


ANOMALY DETECTION

- Over limit speeder flagged at 21.58σ
- Achieving more than 80% detection accuracy
- Over limit speeder is elevated across the entire speed feature block and extreme on high speed gap behaviour



- its measured speed aspects and lateral velocity values fall outside the training distribution



Across all anomaly types, and different regions, results were consistent, achieving Less than 60s pipeline latency (less than 3s per event)

Conclusions

Proven here in simulation. Every street could become a city wide nervous system, sensing the roads everywhere, with no new hardware.